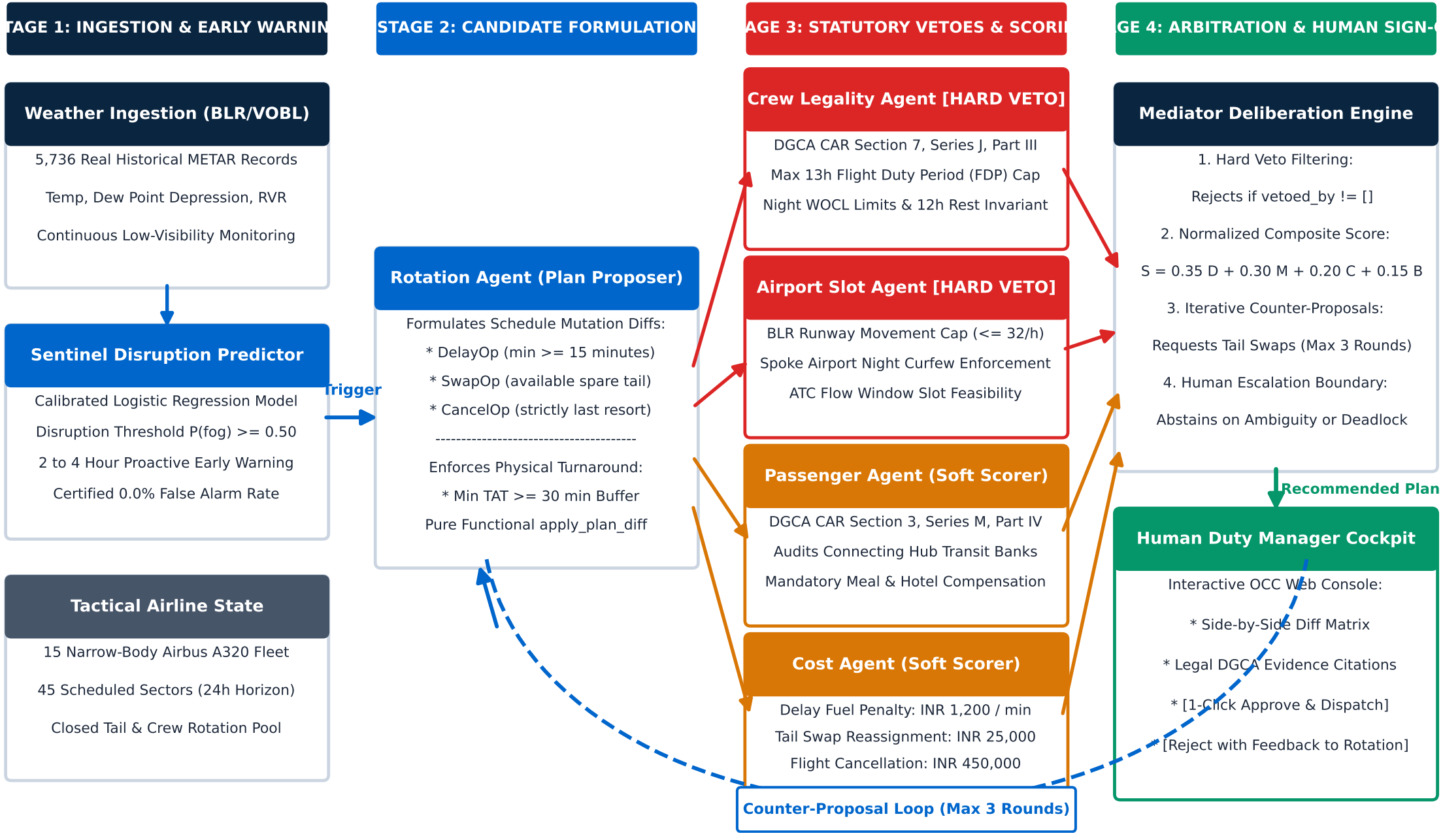
TAGE 1: INGESTION & EARLY WARNIN
STAGE 2: CANDIDATE FORMULATION
AGE 3: STATUTORY VETOES & SCORI
GE 4: ARBITRATION & HUMAN SIGN-
Weather Ingestion (BLR/VOBL)
5,736 Real Historical METAR Records
Temp, Dew Point Depression, RVR
Continuous Low-Visibility Monitoring
Sentinel Disruption Predictor
Calibrated Logistic Regression Model
Disruption Threshold P(fog) >= 0.50
2 to 4 Hour Proactive Early Warning
Certified 0.0% False Alarm Rate
Trigger
Tactical Airline State
15 Narrow-Body Airbus A320 Fleet
45 Scheduled Sectors (24h Horizon)
Closed Tail & Crew Rotation Pool
Rotation Agent (Plan Proposer)
Formulates Schedule Mutation Diffs:
* DelayOp (min >= 15 minutes)
* SwapOp (available spare tail)
* CancelOp (strictly last resort)
Enforces Physical Turnaround:
* Min TAT >= 30 min Buffer
Pure Functional apply_plan_diff
Crew Legality Agent [HARD VETO]
DGCA CAR Section 7, Series J, Part III
Max 13h Flight Duty Period (FDP) Cap
Night WOCL Limits & 12h Rest Invariant
Airport Slot Agent [HARD VETO]
BLR Runway Movement Cap (<= 32/h)
Spoke Airport Night Curfew Enforcement
ATC Flow Window Slot Feasibility
Passenger Agent (Soft Scorer)
DGCA CAR Section 3, Series M, Part IV
Audits Connecting Hub Transit Banks
Mandatory Meal & Hotel Compensation
Cost Agent (Soft Scorer)
Delay Fuel Penalty: INR 1,200 / min
Tail Swap Reassignment: INR 25,000
Flight Cancellation: INR 450,000
Counter-Proposal Loop (Max 3 Rounds)
Mediator Deliberation Engine
1. Hard Veto Filtering:
Rejects if vetoed_by != []
2. Normalized Composite Score:
S = 0.35 D + 0.30 M + 0.20 C + 0.15 B
3. Iterative Counter-Proposals:
Requests Tail Swaps (Max 3 Rounds)
4. Human Escalation Boundary:
Abstains on Ambiguity or Deadlock
Recommended Plan
Human Duty Manager Cockpit
Interactive OCC Web Console:
* Side-by-Side Diff Matrix
* Legal DGCA Evidence Citations
* [1-Click Approve & Dispatch]
* [Reject with Feedback to Rotation]